

Automated IT Onboarding & Offboarding System

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PROBLEM

Manual onboarding and offboarding systems in organizations face numerous issues, such as being time-consuming, prone to errors, and, in certain cases, inconsistent. When onboarding and provisioning accounts face delays, aspects such as employee productivity can be significantly affected, whereas an improper offboarding process can lead to its own set of issues due to carrying the risk of unauthorized access, data breaches, and general compliance violations.

Capstone II Goals

- Develop the Full Web application
- Conduct several usability testing
- Implement the automated workflow
- Deployment of the final system

Capstone 1 progress

During this semester in Capstone 1, the team highlighted on the fundamental research and planning:

- Conducted and identified problem analysis and system requirements.
- Developed the initial user workflows and personas.
- Researched the existing onboarding and offboarding solutions.
- Created the wireframes and early UI prototypes.
- Defined the system technology and architecture.
- Completed the task flows and usability analysis.

Proposed Solution

As a team, we decided to create a web-based IT onboarding and Offboarding system which would automate the entire lifecycle process.

Key Features:

- Role- based access control
- Automated account deactivation and provisioning
- Checklist-based workflows for offboarding and onboarding tasks
- Centralized dashboard for tracking the status of employees
- Notification management for completed and pending tasks.
- Audit logs for compliance and security tracking.

Comparison of systems

Manual Process vs. Automated System

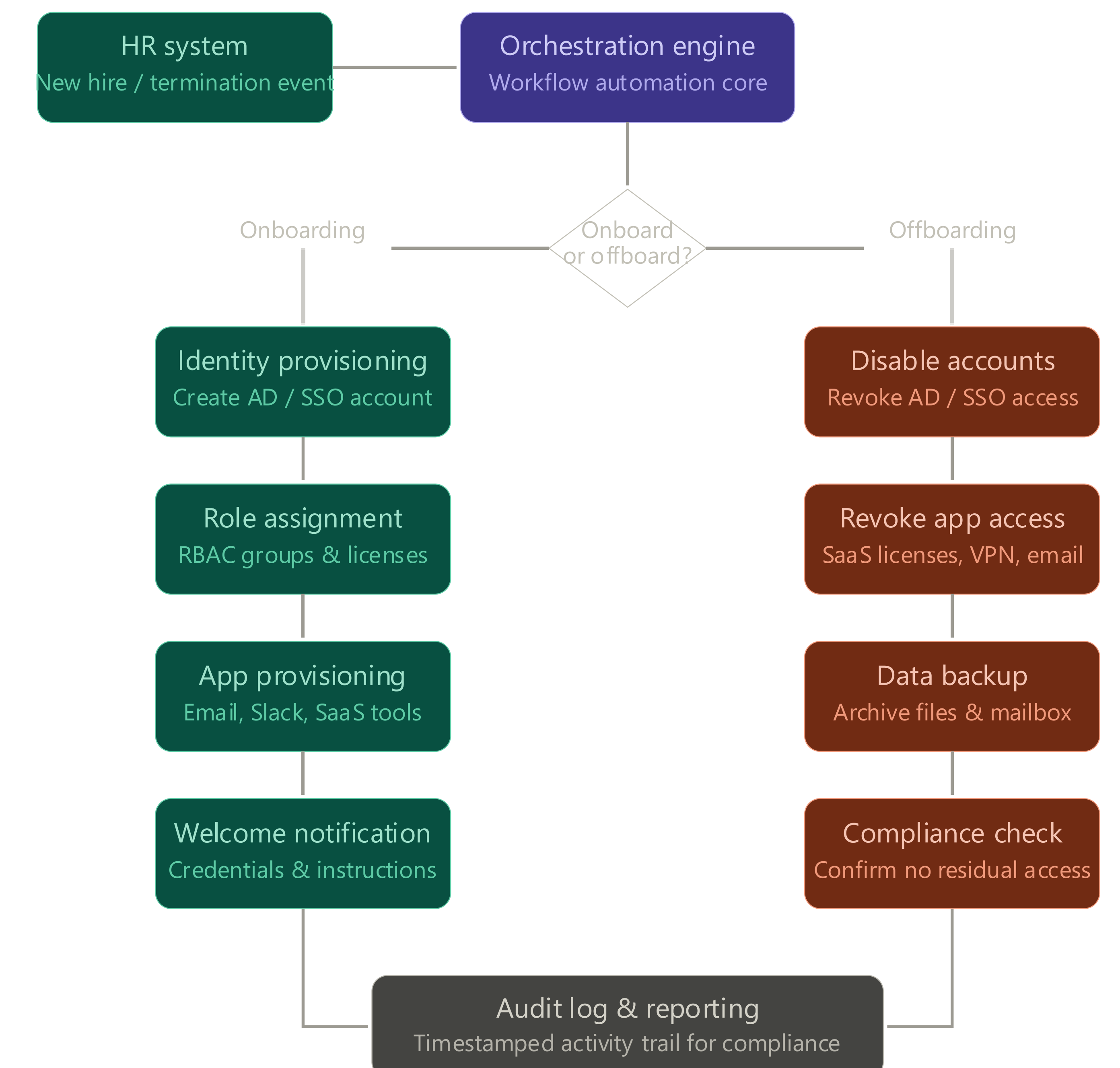
PROBLEM — MANUAL PROCESS

- ✗ **Time-consuming**
Days or weeks to provision accounts and equipment
- ✗ **Error-prone**
Manual steps lead to missed accounts and wrong permissions
- ✗ **Security risk**
Delayed offboarding leaves orphaned accounts and data exposure
- ✗ **Compliance gaps**
No audit trail; inconsistent policy enforcement

SOLUTION — AUTOMATED SYSTEM

- ✓ **Instant provisioning**
Accounts and access ready before the employee's first day
- ✓ **Consistent & repeatable**
Same workflow every time — zero steps skipped
- ✓ **Immediate revocation**
All accounts disabled at termination, no orphaned access
- ✓ **Full audit trail**
Every action timestamped and logged for compliance

System Architecture



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